

FUTURE DRILLING COSTS SIMULATION

TEAM 4

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APRIL 2, 2026

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Overview

Compagnie Pétrolière et Gazière, Inc. (hereafter referred to as “Company”) requested a simulation of possible future values for 2025 drilling costs by modeling annual changes in cost. The Price Analysis group had previously assumed that the arithmetic changes during the 2006-2012 period followed a normal distribution but asked our team to further investigate this assumption. To address this request, our team developed two separate simulations using different distributional assumptions for the 2006-2012 time period. Given our investigation, we recommend that the Company utilize our first simulation that follows the assumption of normality from 2006-2012.

Methodology

The following section describes the structure and preparation of the data for our simulation.

Data Used

Our team utilized a dataset containing estimated drilling costs for crude oil, natural gas, and dry wells from 1991-2007. The dataset included the arithmetic annual change in these costs from one year to the next for each well type. Following the provided recommendations, we excluded the 2007 drilling cost data from our analysis, as it was considered an outlier year.

Data Preparation

To prepare the data for simulation, we treated the costs for oil, gas, and dry wells equally and assumed that a single average cost could be applied across all well types. Prior to restructuring the dataset with this assumption, we calculated the average drilling cost across oil, gas, and dry wells in 2006 to obtain a single starting value for the simulation. With this starting point established, we restructured the dataset to include only the arithmetic changes, with each date and well type combination represented as a single row.

Analysis

This section presents our evaluation of the normality assumption and describes the two simulation scenarios used to model drilling costs.

Testing Normality

To address the Price Analysis group’s request to evaluate whether normality was a valid assumption for the 2006-2012 time period, we created a normal quantile plot and conducted a Shapiro-Wilk test on the arithmetic changes in cost from 1991-2006. The normal quantile plot is shown in Figure 1.

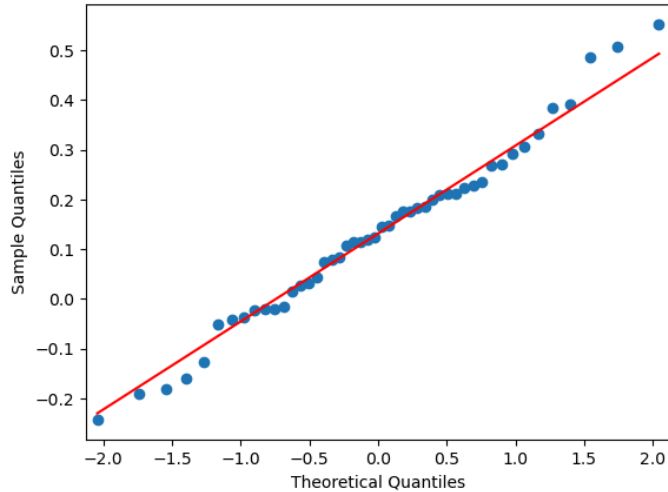


Figure 1: Normal Quantile Plot of Arithmetic Changes from 1991-2006

As shown in Figure 1, the data generally follows the 45-degree reference line, indicating that the distribution is approximately normal. This conclusion is further supported by the Shapiro-Wilk test, which produced a p-value of 0.80. Because this p-value is well above common significance levels, we fail to reject the null hypothesis of normality, suggesting that the distribution of annual cost changes from 1991-2006 is approximately normal. This supports the use of a normal distribution for the simulation of arithmetic changes from 2006-2012. Given this investigation, and the requests of the Price Analysis group, we ran two simulations that are detailed in the following section.

Simulations

In this section, we outline the two simulation scenarios used to model drilling costs for the 2006-2025 time period. The distributional assumptions applied to each time period in each simulation are summarized in Table 1.

Table 1: Outline of Simulations

	Simulation 1	Simulation 2
2006-2012	Normal	Kernel Density Estimate
2012-2015	Uniform with a minimum of -22% and a maximum of -7%	Uniform with a minimum of -22% and a maximum of -7%
2015-2024	Uniform with a minimum of 2% and a maximum of 6%	Uniform with a minimum of 2% and a maximum of 6%
2025	Uniform with a minimum of 2% and a maximum of 6%	Uniform with a minimum of 2% and a maximum of 6%

As shown in Table 1, the two simulations are identical except for the 2006–2012 time period. In Simulation 1, the annual cost changes during this period are modeled using a normal distribution, while Simulation 2 uses a kernel density estimation.

For the 2012-2025 period, the U.S. Energy Information Administration reported that changes in annual drilling costs varied over the period. Between 2012-2015, annual costs decreased by an average of 9.17%, with a maximum decrease of 22% and a minimum decrease of 7%. From 2015-2024, annual costs increased by an average of 5%, with a maximum increase of 6% and a minimum increase of 2%. The U.S. Energy Information Administration expects drilling costs in 2025 to follow the same distribution of increases observed during the 2015–2024 period. These reported distributions are incorporated into the corresponding time periods in both simulations.

Results & Recommendations

This section outlines the results of our analysis and gives recommendations for future decision-making.

Results

After running both simulations, one assuming a normal distribution for the 2006–2012 time period and the other using a kernel density estimate, we found that drilling costs are projected to increase to an average value of approximately \$4.4K across simulations. The probability that costs fall below the initial estimate is approximately 7.85% for Simulation 1 and 9.39% for Simulation 2, indicating that cost reductions are unlikely.

Recommendations

Based on these results, our team recommends using Simulation 1 when simulating possible future values of 2025, as the results are similar to the Simulation 2 as seen in Figure 2.

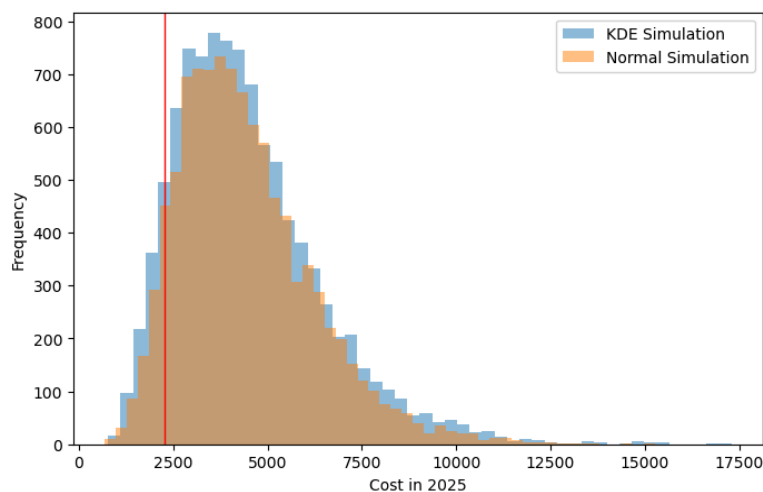


Figure 2: Simulated 2025 Drilling Costs

As shown in Figure 2, both simulations produce very similar results. Given this similarity, Simulation 1 is preferred because it models the 2006–2012 period using a normal distribution, resulting in a simpler model with a lower risk of overfitting.

Conclusion

Based on our evaluation of the normality assumption using historical data, we recommend that the Company utilize Simulation 1, which models the 2006–2012 period using a normal distribution. While this simulation provides a useful estimate of potential drilling costs in 2025, the results should be interpreted as projections rather than exact forecasts.